CHRIST
(DEEMED TO BE UNIVERSITY)
DELHI-NCR, INDIA

NLDL Programming

MCA-575

Assignment – 09

BY

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SUBMITTED TO

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Importing the Libraries:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import accuracy_score
from scipy.spatial.distance import cdist
import math
```

This cell generates a synthetic classification dataset.
- `make_classification` creates 300 samples with 2 informative features and 2 classes.
- The data is stored in `x` (features) and `y` (labels).

```
x , y = make_classification(
    n_samples=300,
    n_features=2,
    n_redundant=0,
    n_informative=2,
    n_clusters_per_class=2,
    n_classes=2,
    random_state=42
)
```

This cell displays the feature matrix `x`.
- Shows the generated feature values for all samples.

```
x
```

```
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```

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```

This cell splits the data and selects RBF centers.
- Splits the dataset into training and testing sets (70% train, 30% test).
- Randomly selects 10 RBF centers from the training data.
- Sets the RBF width parameter `sigma` to 1.0.
- Prints the indices and shape of the chosen centers.

```
# Split the data before using x_train
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3,
random_state=42)
n_centers = 10
rng = np.random.default_rng(0)

center_indices = rng.choice(
    x_train.shape[0],
    n_centers,
    replace=False
)

rbf_centers = x_train[center_indices]
sigma = 1.0  # Standard deviation for RBF

print("center_indices: ",center_indices)
print("rbf_centers shape: ",rbf_centers.shape)
```

```
center_indices:  [170 209 128 103  55   8   3  63  36  15]
rbf_centers shape:  (10, 2)
```

```
scaler = StandardScaler()
x = scaler.fit_transform(x)
```

```
def gaussian_rbf(x_in, centers, sigma):

    dist_sq = cdist(x_in, centers, metric='sqeuclidean')
    phi = np.exp(- dist_sq / (2 * (sigma ** 2)))

    return phi

def rbf_layer(x_in, centers, sigma):
    return gaussian_rbf(x_in, centers, sigma)
```

```
# sanity check
phi_preview = rbf_layer(x_train[:5], rbf_centers, sigma)
print("phi_preview shape (5 samples -> n_centers):", phi_preview.shape)
```

phi_preview shape (5 samples -> n_centers): (5, 10)

```
phi_train = rbf_layer(x_train, rbf_centers, sigma)
```

```
model = LinearRegression()
model.fit(phi_train, y_train)
```

This cell tests the RBF network and computes accuracy.
- Transforms the test data into RBF feature space.
- Uses the trained model to predict continuous outputs, then thresholds them to get class predictions.
- Calculates and prints the test accuracy.

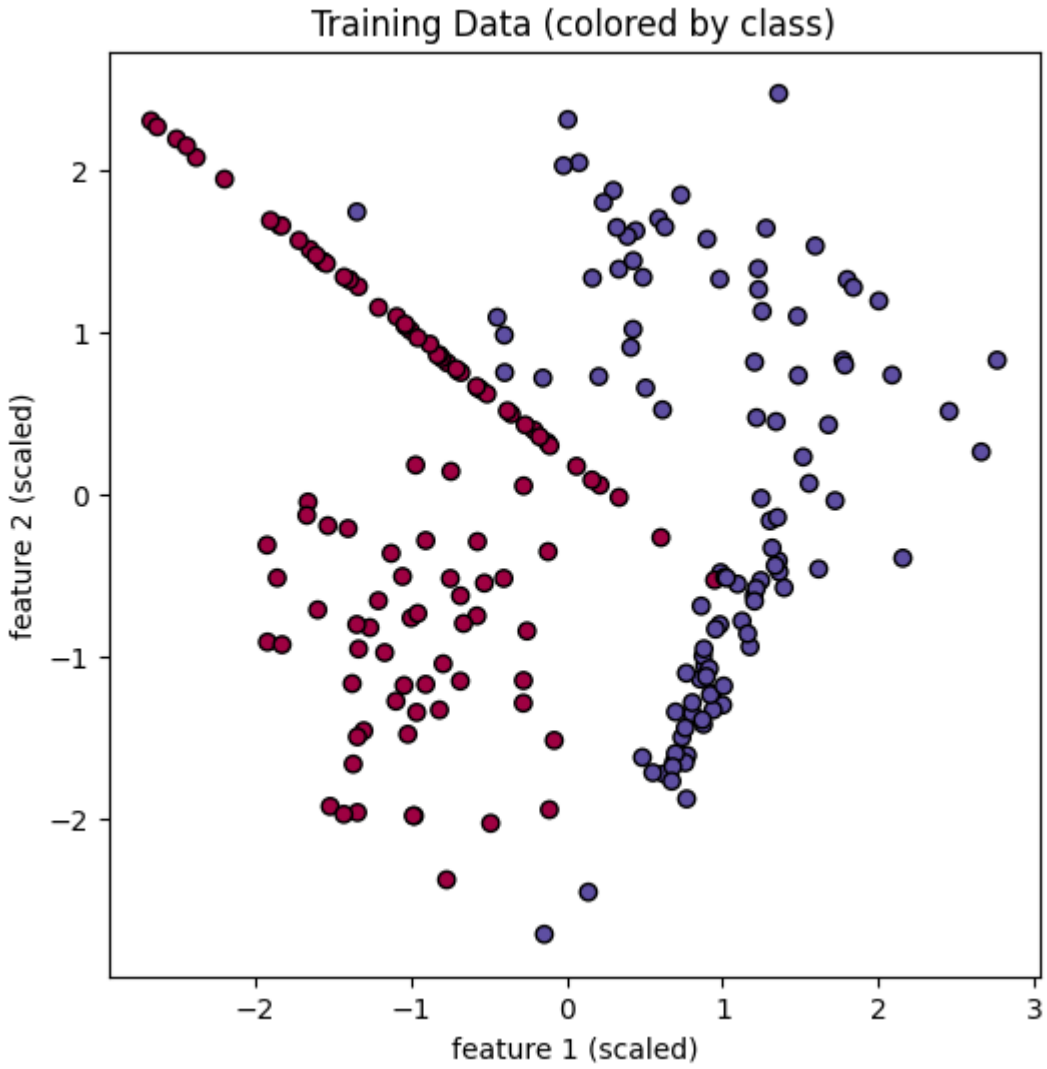
```
phi_test = rbf_layer(x_test, rbf_centers, sigma)
y_pred_cont = model.predict(phi_test)
y_pred_class = (y_pred_cont >= 0.5).astype(int)

acc = accuracy_score(y_test, y_pred_class)
print(f"Test accuracy: {acc * 100:.2f}%")
```

Test accuracy: 95.56%

This cell visualizes the training data.
- Plots the training samples in feature space, colored by their class labels.
- Helps you see the distribution of the two classes.

```
plt.figure(figsize=(6,6))
plt.scatter(
    x_train[:, 0],
    x_train[:, 1],
    c=y_train,
    cmap=plt.cm.Spectral,
    edgecolor="black"
)
plt.title("Training Data (colored by class)")
plt.xlabel("feature 1 (scaled)")
plt.ylabel("feature 2 (scaled)")
plt.show()
```



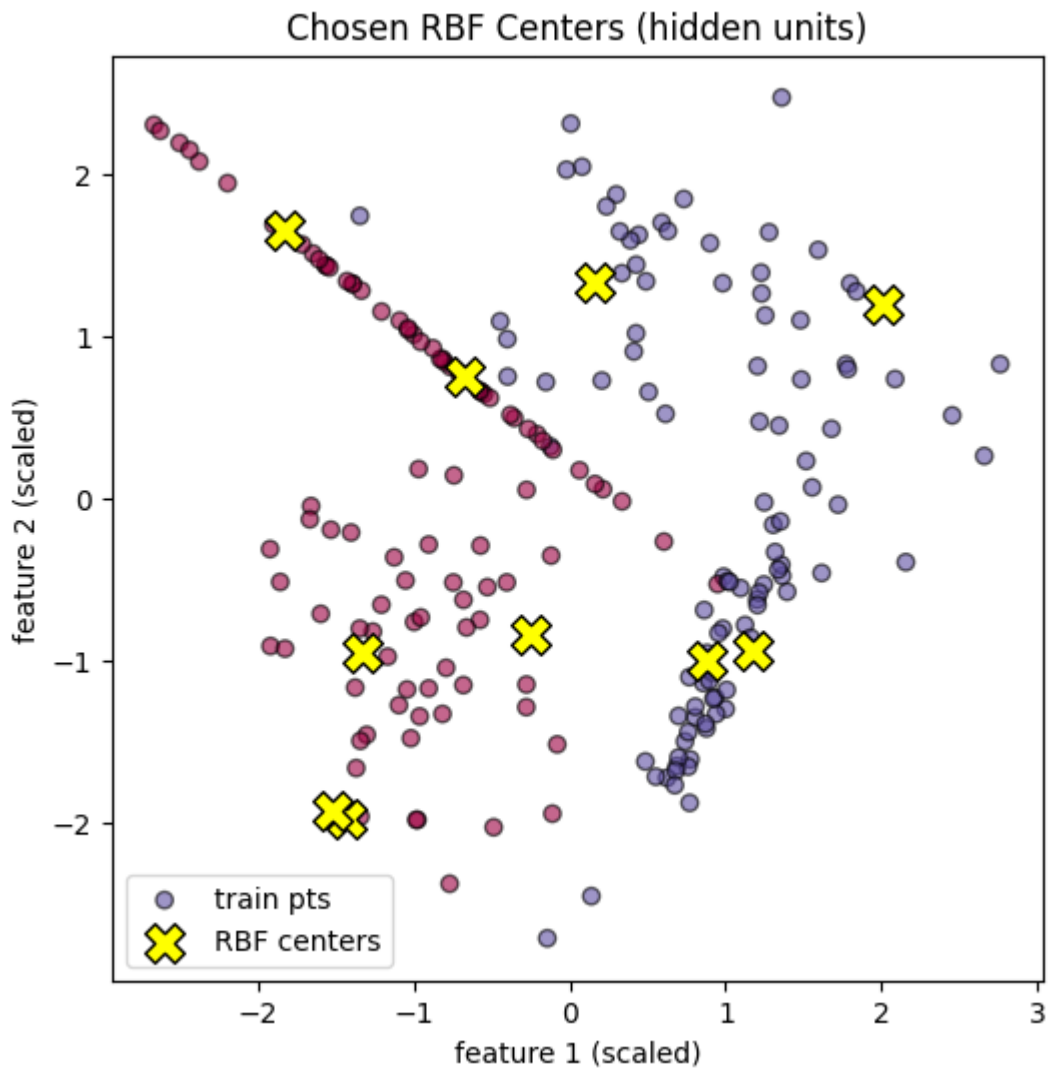
This cell visualizes the chosen RBF centers.
- Plots the training data and highlights the selected RBF centers as yellow X markers.
- Shows which points are used as the basis functions in the hidden layer.

```
plt.figure(figsize=(6,6))

# plot training data
plt.scatter(
    x_train[:, 0],
    x_train[:, 1],
    c=y_train,
    cmap=plt.cm.Spectral,
    edgecolor="black",
    alpha=0.6,
    label="train pts"
)
```

```python
# plot chosen RBF centers
plt.scatter(
    rbf_centers[:, 0],
    rbf_centers[:, 1],
    marker="X",
    s=200,
    c="yellow",
    edgecolor="black",
    label="RBF centers"
)

plt.title("Chosen RBF Centers (hidden units)")
plt.xlabel("feature 1 (scaled)")
plt.ylabel("feature 2 (scaled)")
plt.legend()
plt.show()
```



This cell plots the RBF network's decision boundary.
- Creates a grid of points and predicts their class using the trained RBF network.
- Plots the decision regions, all data points, and RBF centers.
- The plot title shows the test accuracy achieved by the model.

```
x_min, x_max = x[:, 0].min() - 1.0, x[:, 0].max() + 1.0
y_min, y_max = x[:, 1].min() - 1.0, x[:, 1].max() + 1.0

xx, yy = np.meshgrid(
    np.linspace(x_min, x_max, 300),
    np.linspace(y_min, y_max, 300)
)

# flatten grid to list of points, like [[x1,y1],[x2,y2],...]
grid_points = np.c_[xx.ravel(), yy.ravel()]

# RBF features for every grid point
phi_grid = rbf_layer(grid_points, rbf_centers, sigma)

# Predict on grid
z_cont = model.predict(phi_grid)
z_class = (z_cont >= 0.5).astype(int)
z_class = z_class.reshape(xx.shape)  # reshape back to 2D so we can contour
plot

plt.figure(figsize=(6,6))

# background classification regions
plt.contourf(xx, yy, z_class, alpha=0.3, levels=[-1, 0, 1])

# plot all original samples
plt.scatter(
    x[:, 0],
    x[:, 1],
    c=y,
    cmap=plt.cm.Spectral,
    edgecolor="black"
)

# plot RBF centers again
plt.scatter(
    rbf_centers[:, 0],
    rbf_centers[:, 1],
    marker="X",
    s=200,
    c="yellow",
    edgecolor="black",
    label="RBF centers"
```

```
)

plt.title(f"RBF Network Decision Boundary\nTest Accuracy = {acc*100:.2f}%")
plt.xlabel("feature 1 (scaled)")
plt.ylabel("feature 2 (scaled)")
plt.legend()
plt.show()
```